Week 1

Raadhikka Gupta

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1 Foundations in Generative AI

1.1 Generative AI process

- Data Input: AI model is given a large amount of data (images, sounds, text, and numbers)
- Training: AI model analyzes this data and looks for relationship between information. The **neural network** is trained on a dataset on the examples of **outputs**.
- Output: AI model generates new outputs by inputting a **random seed value**. It processes the seed value and generates new output based on the patterns learnt.

1.2 AI-assisted Development

- Building the Foundation
- Solving problems
- Safety Checks
- Documentation
- Refining and polishing

In all cases where AI is used, humans always double-check and ensure that the output produced by the AI is valid and doesn't cause any errors to the whole program.

1.3 Large Language Models

- **Generative AI**: Training the model on TBs of data, (example) then asking it to generate the next word of a sentence/phrase
- **Foundational Model**: It is a type of Generative AI, that are trained on large amounts of data, and it is used as a base for a variety of different tasks
- **Large Language Model**: It is a type of Foundational Model, to understand and process human text. They are used for chatbots as they predict the next word in a sequence based on linguistic patterns. Examples: Claude, Gemini, GPT
- **Vision**: It is a type of Foundational Model, to analyze and understand visual data. It has pre-trained ML models for visual tasks like object classificaton, object detection, and image segmentation.
- **Tuning**: The process of taking a trained model and training it more to make it more specific. Making it personalized for your use case, and giving it custom rules, tones, and data to its core behaviour.
- **Prompt Engineering**: To give instructions to an AI Model to generate an output.

1.3.1 How LLMs are trained?

- **Exposed to dataset**: Given billions of pages from different sources that provides a foundational knowledge needed for training.
- **Pattern Recognition**: Identifies patterns in language like grammar, meaning, and context; by predicting the next word in a sequence of text.
- **Fine-tuning and iteration**: Repeated iterations and applications of algorithms allows it to generate output that are contextually appropriate and human-like.

1.3.2 How do you work with a LLM?

- **Input Prompt**: A question, instruction, or statement.
- **Tokenisation**: Prompt is broken into **tokens**; words or subwords; to interpret and process the input carefully.
- **Contextual understanding**: Uses transformer architecture to analyze the prompt in context, to examine how words relate to each other.
- **Prediction**: Predicts the next most likely token based on the prompt, training data, and statistical patterns learnt.
- **Output generation**: Continues to predict tokens until reaches a stopping point, the output.

1.4 Prompt

- A **prompt** is an input given to the Generative AI system to help it create a specific output.

1.4.1 How to make a prompt work?

- **Clarity**: provides a clear task, good context, and avoids vague instructions.
- **Relevance**: matches the goal to complete, speaks to the correct audience, and uses a format that fits the task.

1.4.2 How to write a prompt?

- **Role**: How the AI should act as and expertise it must use
- **Context**: Background information
- **Task**: What you want the AI to do
- **Output**: What the output should look like (format or structure)
- **Constraints**: Limits, such as length, tone, or required content

2 Large Language Models

2.1 Capability

- **Natural Language Processing**: (To read, understand, and generate human language) This capability lets them analyze words and context, interpret intent, answer question, and even simulate conversations.
- **Text summarization**: Summarize long texts
- **Language translation**: To not only swap words, but also keep the content and nuance of the original message.
- **Content generation**: To generate content